

API and Coding Style Guide

1. Purpose

This guide defines coding conventions, naming standards, and the pull request process for all engineering teams at Nimbus Cloud Systems.

2. Naming Conventions

2.1 Variables and Functions

Use camelCase for variables and functions in JavaScript/TypeScript, and snake_case in Python. Names should be descriptive; avoid single-letter variables except in short-lived loop counters.

2.2 API Endpoints

REST endpoints must use plural nouns and kebab-case, e.g. '/api/customer-orders'. Avoid verbs in endpoint paths; use HTTP methods to express actions (GET, POST, PATCH, DELETE).

2.3 Database Tables

Table names use snake_case and are plural, e.g. 'user_accounts'. Foreign keys should be named '_id', e.g. 'user_id'.

3. Pull Request Process

- Branch names should follow the pattern 'feature/short-description' or 'fix/short-description'.
- PRs should be scoped to a single logical change; avoid bundling unrelated fixes.
- Every PR must include a description of what changed and why, plus a link to the relevant ticket.
- At least one approval is required before merging, per team norms in the Onboarding Guide.
- Squash commits on merge to keep the main branch history clean.

4. Testing Standards

All new features require unit test coverage of at least 80% for new code. Critical paths (payments, authentication) require integration tests in addition to unit tests. Tests must pass in CI before a PR can be merged.

5. Documentation

Public API endpoints must be documented using the OpenAPI specification. Internal libraries should include a README covering purpose, setup, and usage examples.

6. Deprecation Policy

Deprecated API endpoints must remain functional for a minimum of 90 days after a deprecation notice is published, and must return a deprecation warning header during that period.